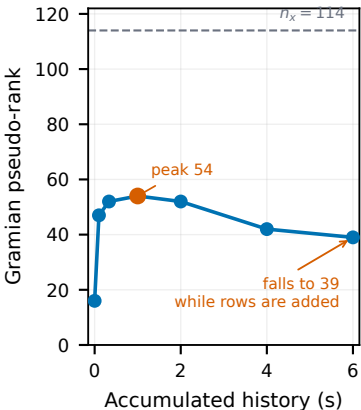
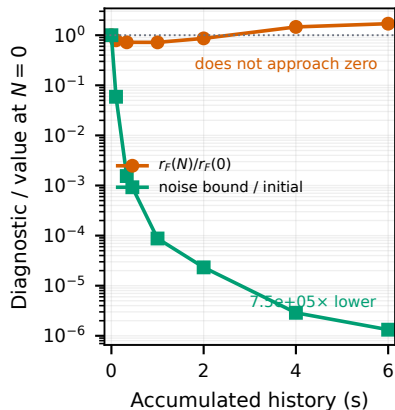


(a) Exact-rank contradiction



(b) Conditioning, not certification



(c) Claim boundary

Do not infer observability from estimator accuracy



Full dynamic state

not established: rank 39/114 and nonmonotonic



Exact voltage-function certificate

not obtained: $r_F(N) > 0$ at every tested horizon



Nominal hidden-voltage recovery

empirically supported: 0.009634% mean HVRE; $\rho = 0.5625$